Dhruv Patel

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Education

University of California, Los Angeles (UCLA) Jun. 2027
B.S. – Mathematics of Computation | Minor – Data Science Engineering *GPA: 3.77/4.00*

Coursework: Machine Learning, Deep Learning & Neural Networks, ML for Finance (Graduate), AI for Scientific Computing (Graduate), NLP, Real Analysis, Probability, Linear Algebra, Data Structures, Algorithms, Numerical Methods, Theory of Computing

Honors: Upsilon Pi Epsilon (CS Honor Society), UCLA Dean's Honors List (3×), Eagle Scout

Organizations: Association for Computing Machinery, UCLA Poker Group, UCLA Cubing Club, Intramural Basketball

Experience

Pacific Life Newport Beach, CA
Data Analytics Intern *Jun. 2026 – Sep. 2026*

- Building a Python pipeline to extract relevant info from medical records, match underwriting guidelines, and classify risk via LLM
- Validating low-risk classifications against the production regression model to reduce forensic underwriter caseload

UCLA Mathematical & Statistical Finance Group Los Angeles, CA
Quantitative Research Assistant *Jun. 2025 – Present*

- Researching ML and AI applications in quantitative finance under Prof. Mihai Cucuringu
- Forecasted daily volume dispersion over a ∼4k stock universe; HAR-based model cut OOS forecast error 18% vs AR(1) baseline
- Built market-neutral long-short equity strategy using volume dispersion; 2.81 Sharpe over 25-year backtest (2000–2024)
- Designed configurable strategy backtesting pipeline, reaching 4× speedup via vectorized NumPy operations and 5D PnL tensor

Smart Circle International Newport Beach, CA
AI Engineer Intern *Sep. 2025 – Jun. 2026*

- Developed a Snowflake Cortex LLM agent for D2D sales territory reporting and zip code-level lead replacement recommendations
- Built a Python pipeline to parse and structure SCORM e-learning content across 10+ formats for downstream AI training workflows
- Aggregated 15+ campaign documents across formats and structures into a centralized knowledge base for LLM ingestion

Analytics Intern *Jun. 2025 – Sep. 2025*

- Built Python churn prediction models (Random Forest & Logistic Regression), achieved 50%+ recall with 85%+ accuracy
- Synthesized 100K+ dispositions into a weekly Tableau report on sales efficiency by office; adopted across 15+ sales campaigns
- Designed office-level metrics evaluating AI-driven recruiting; informed funding allocation decisions across 100+ offices
- Prototyped a Google Places API pipeline to automate B2B lead sourcing as a lower-cost alternative to purchased lead lists

Bruin Sports Analytics Los Angeles, CA
Tennis Data & Scouting Analyst *Oct. 2024 – Jun. 2026*

- Built a transformer model to predict shot type and court location from sequential rally data using PyTorch
- Created 20+ visualizations of serving tendencies used by the UCLA D1 Men's Tennis Team to adjust pre-match strategy
- Standardized end-to-end workflow from match footage to court visualizations, enabling adoption across the scouting team

Projects

LOBSTER News Alpha Agent | *Python (pandas, SciPy, NumPy), Anthropic API*

- Built an LLM agent combining news sentiment with limit-order-book order flow for intraday signals on 10 NASDAQ names
- Ran contamination-controlled evaluation (post-cutoff window, 2,879 events) with Benjamini-Hochberg FDR and bootstrap CIs
- Identified one subgroup surviving FDR: agent beat lexicon baseline on inference-heavy TSLA headlines (IC +0.28, $p < 0.001$)

NBA Timeout Optimization via Hidden Markov Models | *Python (hmmlearn, NumPy, pandas, Scikit-Learn)*

- Fit a 3-state HMM on 250K+ NBA possessions to infer latent momentum regimes from sequential scoring data
- Applied within-team covariate matching across 12K+ pairs to estimate timeout effects net of team quality
- Found timeout impact peaks in Q4 when trailing 6-10 points (+0.14 points/possession)

EEG Motor Imagery Classification | *Python (PyTorch, NumPy, SciPy)*

- Built a data pipeline to resample, bandpass filter, and normalize noisy 22-channel EEG signals across 9 subjects
- Benchmarked five architectures (CNN, LSTM, GRU, TCN, Transformer) under a 36-fold leave-one-subject-out protocol
- Showed compact models (4K parameters) generalize better than 200K-parameter alternatives

Skills

Languages: Python, C++, Java, C, SQL, R, JavaScript, HTML/CSS, LaTeX

Libraries: NumPy, pandas, Scikit-Learn, PyTorch, SciPy, hmmlearn, Optuna, LightGBM, XGBoost, Matplotlib, Seaborn

Tools: Tableau, Excel, Git, Linux, JupyterLab, Google Cloud, Snowflake